

1) What we are building (read first)

Mission

Build a **Procurement Intelligence & Control Agent** — **not a buyer** and **not an ERP replacement**.

It prepares procurement actions, controls workflows, enforces Saudi compliance, keeps audit visibility, and hands approved data to Finance/ERP. AI prepares → Humans approve → Systems execute.

Hard failures (if any happen, the system is wrong)

- Commits spend
 - Skips approvals
 - Hides risk
- Must be prevented by design.

Explicitly out of scope

- Negotiation
 - Contract signing
 - Autonomous PO issuance
 - ERP replacement / duplicating ERP logic
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2) Where this agent sits (system boundaries)

Upstream (input source)

RFQ & Tender Intelligence Agent provides: approved scope packages, quantities/BOQs, assumptions/exclusions, drawings/specs, standards.

Downstream (execution layer)

- Finance Agent
- ERP via adapter (SAP / Oracle / MS Dynamics)

Side systems (read-only)

CRM is visibility + approvals only; not pricing logic, not RFQs, not ERP actions.

3) Inputs: strict contract (intake gate must enforce)

Procurement Agent accepts ONLY:

- Approved Tender ID
- Approved scope packages (by trade)
- Validated quantities / BOQs
- Assumptions & exclusions
- Approved supplier list (AVL)
- Budget & cost codes
- Approval matrix

Rules:

- If approval status $\neq$ APPROVED $\rightarrow$ **STOP** and request approval.
- If data missing $\rightarrow$ ask **ONE question only** (one-question rule).

4) Workflow: mandatory state machine (must implement exactly)

READY $\rightarrow$ RFQ_PREPARED $\rightarrow$ RFQ_SENT $\rightarrow$ QUOTES_RECEIVED $\rightarrow$
NORMALIZED $\rightarrow$ COMPLIANCE_CHECKED $\rightarrow$ DECISION_PACK_READY $\rightarrow$
APPROVED $\rightarrow$ HANDED_TO_FINANCE
Rework loops: REJECTED $\rightarrow$ REQUOTE_REQUESTED $\rightarrow$ RFQ_SENT

Engineering note: implement as a real state machine with allowed transitions + audit logs on every transition.

5) Core modules your team must build

Module A — Supplier Mapping Engine

- Use **Approved Vendor List (AVL) only**
- Filter by trade, geography, certifications, project rules
- Never auto-add suppliers; never RFQ inactive vendors

Module B — RFQ Pack Builder (critical)

For each supplier, create **vendor-scoped RFQ pack**:

- Only relevant drawings/specs/BOQ lines/standards
- Generate a manifest listing included files
- Hard rule: **zero information leakage between suppliers** (must be tested)

Module C — RFQ Issuance & Tracking

- Issue RFQs via email / portal adapter
- Track sent/opened/responded/overdue
- Reminders + overdue escalation to humans
- AI must not discuss pricing or negotiate

Module D — Quote Ingestion & Extraction

Store original quote read-only and extract:

- Line items, qty, unit rates, lead time, exclusions, payment terms, VAT treatment
Each extracted field must include:
- Confidence score
- Source reference (doc + page)

Module E — Quote Normalisation (apple-to-apple)

Canonical structure:

`line_id | description | uom | qty | unit_rate | total`

Status per line: MATCHED / PARTIAL / MISSING / EXTRA

Rules:

- Align quantities + flag gaps/exclusions/deviations
- Never “fix” quantities automatically

Module F — Saudi Compliance Engine (deterministic rules)

Must enforce:

- VAT/ZATCA registration + correct VAT handling
- Supplier certifications
- Local content rules (where applicable)
- Owner/project packs: Aramco / SEC / RCJY / NEOM / Red Sea / Diriyah
Output: passed/failed + severity (HIGH blocks, MEDIUM review)
HIGH severity = cannot proceed

Module G — Fraud & Risk Controls

- Bank details change detection: mismatch vs vendor master → HARD BLOCK + human verification
- Procurement risk score per supplier: compliance, delivery, scope, commercial → LOW/MED/HIGH

Module H — Approval Orchestration

Configurable layers (typical):

- Engineering confirmation
- Procurement approval
- Commercial approval

- Project Director approval
No approval → no Finance handover

Module I — Handover to Finance (strict payload)

Only after approvals:

Generate **PO Readiness Payload** including selected supplier, approved pricing/terms, compliance status, audit references. Finance handles PO creation, matching, payments. Use the JSON schemas provided (Proc→Finance PO payload, Finance→Proc status update, optional delivery milestone).

Module J — Audit & Security

- Secure object storage (S3/Azure Blob)
 - Originals read-only
 - Log every action: who/when/what/why + source reference
 - Saudi data residency respected where required
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6) Human Review UI (required for MVP)

Must include:

- Quote comparison table
 - Compliance flags
 - Risk scores
 - Approval buttons
 - Audit trail view
 - Export decision pack (PDF/Excel)
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7) Integrations (what to build now)

Day-1 connectors (MVP)

- Email sending + tracking (for RFQs)
- Portal adapter interface (pluggable)
- Secure storage (S3 / Azure Blob)
- ERP Adapter Layer (SAP / Oracle / Dynamics) — **adapter only, don't rebuild ERP**

Saudi market compliance toggles

Project config must support enabling rule packs:

Aramco / SEC / RCJY / NEOM / Red Sea / Diriyah

8) Testing / acceptance criteria (non-negotiable)

System is not “done” unless:

- RFQ packs are vendor-scoped and leak-proof
 - Quotes normalized + traceable
 - VAT non-compliance blocks progression
 - No finance handover without approvals
 - Audit trail shows every step
 - Re-quote + addendum flows work
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9) Jira-style build plan (copy/paste for tickets)

Your spec already includes epics + stories. Use these directly as the baseline backlog:

Core Service → RFQ Packaging → Quote Intake/Normalization → Compliance Engine → Approvals → Finance Handover → Audit/Observability.
